

Notes: Gu, Russell & Stringham (RES, 2026)

Counterfactual Identification and Latent Space Enumeration in Discrete Outcome Models

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1 Big picture

- **Question.** How can researchers sharply bound counterfactual parameters in discrete outcome models with endogenous regressors and multidimensional unobserved heterogeneity, without making parametric distributional assumptions?
- **Core obstacle.** Standard counterfactual analysis requires reasoning about a latent variable distribution $P_{U|X}$ and counterfactual outcome distributions. These are infinite-dimensional objects when the unobservables are continuous or multidimensional.
- **Main theoretical contribution.** When observed outcomes and covariates are discrete, the infinite-dimensional latent variable distribution can be replaced by a finite-dimensional distribution supported on representative points from cells of a latent-space hyperplane arrangement.
- **Computational contribution.** The paper develops a new efficient cell enumeration algorithm for hyperplane arrangements. It enumerates cells exactly rather than approximately and is much faster than previous enumeration methods in the relevant designs.
- **Counterfactual computation.** Once latent space is discretised into finitely many cells, sharp bounds on counterfactual functionals can be computed by solving linear programming problems.
- **Why discrete covariates matter.** Discrete X implies that the model inequalities generate a finite collection of hyperplanes. These hyperplanes partition the latent space into finitely many cells. Within a cell, all latent draws have the same implications for observed and counterfactual outcomes.
- **Why multidimensional latent variables matter.** Many empirical models involve multiple latent shocks: random coefficients, product-specific taste shocks, payoff shocks in games, heterogeneous choice sets, or dynamic unobservables. Grid methods become infeasible in such settings.
- **Linearity-in-parameters extension.** When the model indexes are linear in structural parameters, the paper also enumerates equivalence classes in the parameter space. This avoids solving the same linear programs repeatedly for observationally equivalent parameter values.
- **Applications.** The paper applies the method to a mobile-phone choice example with heterogeneous choice sets and an airline entry game. The applications show how additional assumptions such as independence, sign restrictions, fixed coefficients, and equilibrium restrictions tighten bounds.
- **Economic takeaway.** The paper transforms a difficult nonparametric partial-identification problem into a finite but potentially large linear programming problem. The key insight is that, for discrete outcomes, only the cell of the latent variable matters, not the exact point inside the cell.

2 Model primitives and identified sets

2.1 General discrete outcome model

- The observed outcome vector is $Y \in \mathcal{Y}$ and the observed covariate vector is $X \in \mathcal{X}$.
- Both $\mathcal{Y}$ and $\mathcal{X}$ are finite sets.
- The latent variable is $U \in \mathcal{U}$, which can be continuous and multidimensional.
- The structural parameter is $\theta \in \Theta$.
- For every candidate outcome y , covariate x , latent draw u , and parameter θ , the model gives a set of inequalities that determine whether y is compatible with the model.

The paper expresses the model through a random set of feasible outcomes:

$$\mathcal{Y}(x, u, \theta) = \{y \in \mathcal{Y} : g_{j,k}(y, x, u, \theta) \geq 0 \text{ for all relevant } j, k\}. \quad (1)$$

Interpretation: The model can be incomplete or incoherent. Incomplete means more than one outcome may be feasible; incoherent means no outcome may be feasible. The random-set representation allows the paper to include games with multiple equilibria, discrete choice models with endogeneity, and dynamic binary response models in one framework.

2.2 Selection mechanism

A selection mechanism is a conditional distribution over observed outcomes given covariates and latent variables:

$$P_{Y|X,U}(y|x, u). \quad (2)$$

It must assign probability one to outcomes that are feasible under the model:

$$P_{Y|X,U}(Y \in \mathcal{Y}(x, U, \theta) \mid x, U) = 1. \quad (3)$$

Interpretation: The selection mechanism is needed because the model may not select a unique outcome. This is especially important in games, where multiple equilibria can arise, and in models where the theory leaves some choices unspecified.

2.3 Identified set of structures

The paper defines the identified set of structures as the set of pairs $(P_{U|X}, \theta)$ for which there exists a valid selection mechanism matching the observed conditional distribution of Y given X :

$$P_{Y|X}(B|x) = \int_{\mathcal{U}} \sum_{y \in B} \mathbf{1}\{y \in \mathcal{Y}(x, u, \theta)\} P_{Y|X,U}(y|x, u) dP_{U|X}, \quad (4)$$

for every $B \subseteq \mathcal{Y}$ and every $x \in \mathcal{X}$.

Interpretation: Identification is cast as an existence problem. A structure is observationally equivalent to the data if some latent distribution and some admissible selection mechanism can rationalise the observed conditional choice probabilities.

2.4 Counterfactuals

A counterfactual is indexed by a parameter γ and produces counterfactual outcomes Y^γ . The counterfactual outcome set is

$$\mathcal{Y}^\gamma(x, u, \theta). \quad (5)$$

A counterfactual conditional distribution belongs to the identified set if there exists an observationally equivalent structure and a counterfactual selection mechanism satisfying the corresponding feasibility constraints.

Interpretation: The counterfactual is not assumed point identified. Instead, the paper derives sharp bounds on functionals of the counterfactual distribution, such as probabilities of choosing a product, entry probabilities, or changes in choice probabilities under a policy intervention.

2.5 Counterfactual functional of interest

A large class of counterfactual parameters can be written as

$$\Psi_x(P_{Y^\gamma|X}) = \sum_{y^\gamma \in \mathcal{Y}^*} \phi_\gamma(y^\gamma, x) P_{Y^\gamma|X}(y^\gamma|x). \quad (6)$$

Interpretation: This includes counterfactual choice probabilities, average treatment effects, and differences in counterfactual outcome probabilities. The objective is linear in the counterfactual distribution, which is why linear programming becomes possible.

3 Finite latent-space representation

3.1 Latent cells

For a fixed θ , the functions $g_{j,k}(y, x, u, \theta)$ define hyperplanes in the latent space. These hyperplanes partition $\mathcal{U}$ into finitely many cells:

$$\mathcal{U}_{\theta,1}, \dots, \mathcal{U}_{\theta,L_\theta}. \quad (7)$$

Within each cell, the sign of every relevant index is constant.

Interpretation: If two latent draws lie in the same cell, then they generate the same feasible observed and counterfactual outcome sets. Therefore, the exact distribution within a cell is irrelevant for identification.

3.2 Witness points

For each cell $\mathcal{U}_{\theta,\ell}$, choose one representative point $u_{\theta,\ell} \in \mathcal{U}_{\theta,\ell}$. These representative points are called witness points.

Interpretation: The witness point is not an approximation. It is an exact representative of the entire cell from the perspective of the model's inequality signs.

3.3 Theorem 2.1: finite-dimensional equivalence

The paper's first main theorem says that the infinite-dimensional latent distribution can be replaced, without loss for identification, by a distribution supported on witness points:

$$\{u_{\theta,1}, \dots, u_{\theta,L_\theta}\}. \quad (8)$$

In particular, any admissible counterfactual distribution generated by a continuous latent distribution can also be generated by a finite distribution over these cells, and any finite distribution over these cells corresponds to an admissible latent distribution.

Interpretation: The theorem is the core identification result. It says that the researcher does not need to search over all possible latent distributions. It is enough to search over the probability mass assigned to finitely many cells.

3.4 Why the theorem is sharp

The theorem is sharp because it preserves exactly the same possible distributions over observed and counterfactual outcomes. It does not discretise the latent variable space by a grid. Instead, it exploits the exact geometry of the model.

Interpretation: Grid methods approximate continuous latent variables and can miss cells or create artificial cells. The cell method is exact because it follows the hyperplane arrangement induced by the model.

4 Linear programming representation of bounds

4.1 Cell probabilities

Let $\pi_{x,\ell}$ denote the probability mass assigned to cell ℓ conditional on $X = x$:

$$\pi_{x,\ell} = P(U \in \mathcal{U}_{\theta,\ell} | X = x). \quad (9)$$

Let $q_{x,\ell}(y)$ and $q_{x,\ell}^\gamma(y^\gamma)$ denote selection probabilities over observed and counterfactual outcomes at the witness point.

Interpretation: The linear programs choose latent cell masses and selection probabilities that match the data and satisfy counterfactual feasibility.

4.2 Observed-data constraints

For every observed covariate value x and outcome set $B \subseteq \mathcal{Y}$, the observed distribution must be matched:

$$P_{Y|X}(B|x) = \sum_{\ell} \pi_{x,\ell} \sum_{y \in B} \mathbf{1}\{y \in \mathcal{Y}(x, u_{\theta,\ell}, \theta)\} q_{x,\ell}(y). \quad (10)$$

Interpretation: These constraints are linear because the feasible outcome indicators are fixed once witness points are chosen.

4.3 Counterfactual constraints

The counterfactual outcome distribution must satisfy

$$P_{Y^\gamma|X}(B|x) = \sum_{\ell} \pi_{x,\ell} \sum_{y^\gamma \in B} \mathbf{1}\{y^\gamma \in \mathcal{Y}^\gamma(x, u_{\theta,\ell}, \theta)\} q_{x,\ell}^\gamma(y^\gamma). \quad (11)$$

Interpretation: The same latent cell masses used to rationalise the observed data must also generate the counterfactual distribution. This is what links observed and counterfactual choices.

4.4 Theorem 2.2: sharp bounds via linear programs

The lower and upper sharp bounds on Ψ_x solve two linear programs:

$$\underline{\Psi}_x(\theta) = \min \Psi_x(P_{Y^\gamma|X}), \quad (12)$$

$$\overline{\Psi}_x(\theta) = \max \Psi_x(P_{Y^\gamma|X}), \quad (13)$$

subject to the observed-data constraints, counterfactual feasibility constraints, nonnegativity, and probability-sum constraints.

Interpretation: Theorem 2.2 turns counterfactual identification into computation. Once the cells are enumerated, the bounds are the solutions to finite-dimensional linear programs.

4.5 Adding assumptions

Additional assumptions enter as additional linear constraints whenever they can be expressed in terms of cell probabilities and selections. Examples include:

- independence or partial independence restrictions;
- monotonicity restrictions;
- sign restrictions on coefficients;
- fixed-coefficient restrictions;
- median-zero or quantile restrictions;
- equilibrium selection restrictions in games.

Interpretation: The method is modular. The baseline linear programs give nonparametric sharp bounds, and assumptions tighten the bounds by shrinking the feasible set.

5 Cell enumeration algorithms

5.1 Why enumeration is hard

A model with many outcomes, covariate values, and latent dimensions can generate a large number of hyperplanes. A naive enumeration method may require solving many linear programs just to find cells.

Interpretation: The computational bottleneck is not the idea of finite cells, but finding all cells efficiently.

5.2 Algorithm 1: two-dimensional arrangements

For $d_u = 2$, the paper enumerates cells by ordering lines according to their angle and constructing witness points from adjacent lines on the unit circle.

Interpretation: The two-dimensional case is simple because lines through the origin partition the circle into arcs, and each arc corresponds to a cell.

5.3 Algorithm 2: higher-dimensional arrangements

For $d_u > 2$, the algorithm recursively adds hyperplanes. A newly added hyperplane intersects existing hyperplanes and induces a lower-dimensional arrangement on the new hyperplane. Witness points for newly created cells are constructed by perturbing lower-dimensional witness points away from the new hyperplane.

Interpretation: This recursion avoids repeatedly solving feasibility problems. It uses the geometry of hyperplanes to construct witnesses directly.

5.4 Degenerate arrangements

The algorithm handles degeneracy, such as overlapping or non-general-position hyperplanes, by using sign vectors and dropping duplicates.

Interpretation: This is essential for economic models because hyperplanes often overlap or repeat when covariates are discrete or restrictions create redundant inequalities.

5.5 Profiling algorithm for linear-in-parameters models

When model indices are linear in parameters, the same logic is applied to the joint space (u, θ) . The profiling algorithm identifies hyperplanes in the parameter space that define equivalence classes.

Interpretation: Instead of solving linear programs for every possible θ , the researcher solves them for representative parameter values. Equivalent parameter values induce the same cell structure and hence the same feasible counterfactual distributions.

5.6 Full computational procedure

The complete procedure is:

1. Use the profiling algorithm to obtain hyperplanes that partition parameter space.
2. Enumerate representative parameter values from those equivalence classes.
3. For each representative parameter, enumerate latent-space cells.
4. Construct cell-level feasibility indicators for observed and counterfactual outcomes.
5. Solve lower- and upper-bound linear programs.

6. Optionally stitch cells to reduce the dimension of the linear programs.

Interpretation: The algorithm converts a nonparametric counterfactual problem into a finite sequence of exact enumeration and linear programming steps.

6 Applications

6.1 Mobile phone choice application

The first application uses an experimental mobile phone choice dataset previously analysed by Keane and Wasi. Each individual chooses one of three phones or an outside option:

$$Y_i = \arg \max_{j \in J_i} \xi_{ij}, \quad \xi_{ij} = P_{ij} \beta_i^P + W_j \beta_i^W + U_{ij}. \quad (14)$$

- P_{ij} is the price of phone j for consumer i .
- W_j indicates whether phone j has WiFi.
- J_i is the consumer's choice set.
- The outside option has zero price and zero WiFi.

Interpretation: The model allows random coefficients and product-specific taste shocks, creating a multidimensional latent variable.

6.2 Mobile phone counterfactuals

The paper studies two counterfactuals:

- **P1:** reduce the price of phone 1 from 250 AUD to 50 AUD, holding other prices fixed.
- **P2:** eliminate phone 3 from the choice set, changing the choice set from $\{0, 1, 2, 3\}$ to $\{0, 1, 2\}$.

Additional assumptions considered include:

- partial independence (I);
- median-zero restrictions (MZ);
- fixed coefficients (FC);
- price sign restrictions ($SR1$);
- WiFi sign restrictions ($SR2$).

Interpretation: The application illustrates how researchers can add economically motivated assumptions and observe how they tighten bounds.

6.3 Mobile phone findings

- For the price-reduction counterfactual P1, bounds are wide under weak assumptions but tighten when sign restrictions and median-zero restrictions are imposed.
- The tightest nonempty interval reported is approximately $[0.00, 0.48]$.
- For the choice-set counterfactual P2, bounds remain $[0.00, 0.29]$ under many combinations of assumptions, matching the nonparametric upper bound implied by observed shares.
- The results show that price counterfactuals can be informative under weak assumptions, while choice-set counterfactuals with heterogeneous choice sets can remain hard to identify.

Interpretation: Removing an option from a heterogeneous choice set can be intrinsically difficult to bound without stronger restrictions because the analyst does not observe which choices each consumer actually considered.

6.4 Airline entry game application

The second application uses airline route-entry data from Kline and Tamer. The model studies entry by low-cost carriers (LCs) and other airlines (OAs):

$$Y_i^{LC} = \mathbf{1}\{\beta_1^{LC} M_i^{LC} + \beta_2^{LC} S_i + \delta_i^{LC} Y_i^{OA} \geq \epsilon_i^{LC}\}, \quad (15)$$

$$Y_i^{OA} = \mathbf{1}\{\beta_1^{OA} M_i^{OA} + \beta_2^{OA} S_i + \delta_i^{OA} Y_i^{LC} \geq \epsilon_i^{OA}\}. \quad (16)$$

- Markets are origin-destination pairs.
- Entry decisions are simultaneous.
- Multiple equilibria can arise.
- The paper studies counterfactuals based on moving airport presence and changing market size variables.

Interpretation: The airline game application shows that the framework handles strategic interaction and multiple equilibria, not just single-agent discrete choice.

6.5 Airline findings

- Bounds depend strongly on equilibrium restrictions.
- Some equilibrium restrictions have little effect for some counterfactuals but substantial effects for others.
- Computationally, the method handles large cell counts and high-dimensional linear programs, but memory and time requirements can be substantial.

Interpretation: The application illustrates the practical value of the algorithm and the substantive importance of assumptions in partially identified games.

7 Tables, figures, and algorithms

7.1 Table 1 and Figure 1: model components and latent cells

Read:

- Table 1 shows how the unified model nests discrete choice, games, and dynamic binary response models.
- Figure 1 illustrates a hyperplane arrangement in latent space and the sign vectors associated with cells.
- The key visual point is that the latent space is divided into finitely many cells even when the latent variable is continuous.

Economic message: The same geometric object - a hyperplane arrangement - connects identification theory and computation.

TABLE 1
The specification of the components of Assumption 2.1 for Examples 1, 2 and 3

Example	d_y	$\mathcal{Y}$	$f_k(w_{1,k}, \dots, w_{J,k})$	$g_{j,k}(y, x, u, \theta)$
Discrete Choice (Full Choice Set)	1	$\{1, \dots, J\}$	$\min_{j' \in \mathcal{J}} \{j' \cdot \prod_{j \in \mathcal{J} \setminus \{j'\}} w_{j,k}\}$	$(x_{j'} - x_j)^\top \theta + u_{j'} - u_j$
Game	K	$\{0, 1\}^K$	$w_{1,k}$	$x_k^\top \beta_k + y_{(-k)}^\top \delta_k - u_k$
Dynamic Binary Response	T	$\{0, 1\}^T$	$w_{1,k}$	$x_k^\top \beta + y_{k-1}^\top \delta + u_k$

Table 1: model components for examples

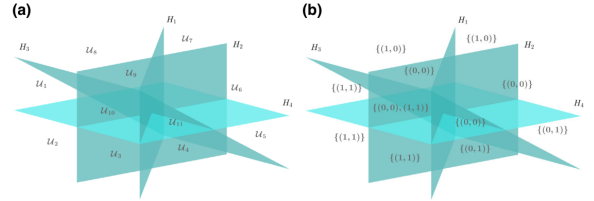


FIGURE 1
(a) The figure illustrates 11 of 14 cells induced by the hyperplanes H_1, H_2, H_3 and H_4 . Cells $\mathcal{U}_{12}, \mathcal{U}_{13}$ and $\mathcal{U}_{14}$ are hidden behind the figure and (b) The figure displays the Y -level sets associated with each cell in the arrangement from Figure 1(a).

Figure 1: hyperplane cells and sign vectors

7.2 Figure 2 and Algorithms 1–2: cell enumeration

Read:

- Figure 2 explains how the algorithm adds a new hyperplane and constructs witness points in newly created cells.
- Algorithm 1 handles the two-dimensional base case.
- Algorithm 2 recursively handles higher-dimensional arrangements.
- The method avoids solving a feasibility problem for every candidate cell.

Economic message: This is the computational engine of the paper. It makes exact nonparametric counterfactual analysis feasible in models with multidimensional unobservables.

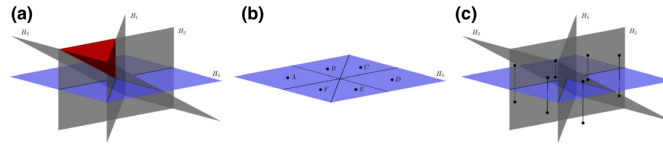


FIGURE 2

Figure (a) illustrates an arrangement consisting of $n_h = 4$ hyperplanes in $d_h = 3$ dimensional space. The newly added hyperplane H_4 is in blue. Figure (a) also highlights a cell which is not crossed by H_4 . In Figure (b), the lower-dimensional arrangement is represented by the three black lines formed by the intersection of H_1, H_2 and H_3 with H_4 . Figure (b) also shows the six witness points found in the lower-dimensional arrangement. Figure (c) illustrates the perturbation procedure to move witness points from the lower-dimensional arrangement into the newly crossed

Algorithm 1: ($d_u = 2$)

input : $n \times 2$ matrix H , where $Hu = 0$ represents a linear arrangement.
output: Matrix of witness points W and the associated sign vectors S .

- Drop duplicated (up to a constant) rows of H and call the resulting matrix H^* .
- Let n_u be the number of rows in H^* . If $n_u = 1$, proceed to step 3. Otherwise, proceed to step 4.
- If $n_u = 1$:
 - Pick a random point w_1 in $\mathbb{R}^{d_u}$.
 - If $\langle w_1^*, w_1 \rangle < 0$, set $w_1 \leftarrow -w_1$.
 - Set $W^+ = w_1$. Proceed to step 5.
- If $n_u > 1$:
 - Set $p_i = \frac{1}{\|h_i^*\|} h_i^* \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ for $i = 1, 2, \dots, n_u$ of H^* , where h_i^* is the i th row of H^* .
 - For each i , if $\langle h_i^*, p_i \rangle$ is negative, set $p_i = -p_i$.
 - Order the points p_i by $\langle h_i^*, p_i \rangle$ (increasing), and let the ordered points be q_j , with $j = 1, 2, \dots, n_u$.
 - For $j \leq n_u - 1$, let $w_j = \frac{1}{2}(q_j + q_{j+1})$.
 - Let $w_{n_u} = \frac{1}{2}(q_{n_u} - q_1)$.
 - Let W^+ be the $2 \times n_u$ matrix with points w_j as columns.
- Let W be the matrix composed of the columns of W^+ and the columns of $-W^+$. Let S be the matrix of signs of HW . Return W and S .

Algorithm 1: base case $d_u = 2$

Algorithm 2: ($d_u > 2$)

input : An $n_h \times d_u$ matrix H , where $Hu = 0$ represents a linear arrangement.
output: A matrix of witness points W and the matrix of sign vectors S .

- Pick a random point w_1 in $\mathbb{R}^{d_u}$. If $\langle h_1, w_1 \rangle < 0$, set $w_1 \leftarrow -w_1$.
- Store w_1 as a column in the matrix W^+ , and initialise S^+ as an $n_h \times 1$ zero matrix, then set $S_{11} = 1$.
- For $i = 2, 3, \dots, n_h$:
 - Let $s_i = \text{sgn}(h_i^T W^+)$, and store s_i as the i th row of S^+ .
 - If h_i is a duplicated row (up to a constant) set $i \leftarrow i + 1$ and go to 3(a).
 - Let $H_p = H_{1:(i-1)} N(h_i)$, where $N(\cdot)$ denotes a $d_u \times (d_u - 1)$ orthonormal matrix with columns given by a basis of the orthogonal complement of h_i .
 - If $d_u - 1 > 2$, call Algorithm 2 (this algorithm) recursively on H_p . Otherwise, call Algorithm 1. Let W_p denote the returned witness points.
 - Set $W^* = N(h_i) W_p$, where $N(\cdot)$ denotes a $d_u \times (d_u - 1)$ matrix with columns given by a basis of the orthogonal complement of h_i .
 - Let J_i be the number of columns in W^* . For $k = 1, 2, \dots, J_i$:
 - Let m_k be some number in $(0, \min_{j=1, \dots, J_i-1} \left| \frac{\langle h_j, w_k^* \rangle}{\langle h_j, h_i \rangle} \right|)$, where w_k^* is the k th column of W^* .
 - Add and subtract $m_k h_i$ to the k th column of W^* to create two new vectors $\tilde{w}_1$ and $\tilde{w}_2$.
 - Find the sign vectors $\tilde{s}_1 := H_{1:(i-1)} \tilde{w}_1$ and $\tilde{s}_2 := H_{1:(i-1)} \tilde{w}_2$.
 - (Optional) Determine the column in W^+ whose sign vector matches either $\tilde{s}_1$ or $\tilde{s}_2$, and discard it from W^+ . Also discard its sign vector from S^+ .
 - Redefine $W^+ \leftarrow \text{concatenate}(W^+, \tilde{w}_1, \tilde{w}_2)$ (column-wise) and $S^+ \leftarrow \text{concatenate}(S^+, \tilde{s}_1, \tilde{s}_2)$ (column-wise).
- Let W be the matrix composed of the columns of W^+ and $-W^+$, and let S be the matrix composed of the columns of S^+ and $-S^+$. If step 3(f)(iv) was skipped, then drop duplicate column vectors in S and the corresponding column vectors in W , and return W and S ; else return W and S .

Algorithm 2: recursive enumeration for $d_u > 2$

7.3 Table 2: algorithm runtime comparison

Read:

- Table 2 compares the proposed algorithm, denoted GRS, with algorithms from Gu and Koenker and Rada and Cerny.
- In low dimensions and moderate numbers of hyperplanes, GRS is often orders of magnitude faster.
- In the $d_u = 4$ general-position design, alternative algorithms become infeasible for larger numbers of hyperplanes, while GRS remains feasible.
- In the degenerate design, the proposed method also performs well as dimensionality and hyperplane count grow.

Economic message: Runtime matters because counterfactual bounds require cell enumeration repeatedly. The new algorithm turns otherwise infeasible identified-set computations into practical ones.

TABLE 2
Computation time in seconds of the proposed algorithm (denoted GRS) compared to the algorithms of Gu and Koenker (2020) (GK) and Rada and Černý (2018) (RC), which were implemented using the RCBR package in R

Design 1 (general position)							
n_h	$d_u = 2$			$d_u = 3$		$d_u = 4$	
	GRS	GK	RC	GRS	GK/RC	GRS	GK/RC
25	< 0.01	0.03	0.17	< 0.01	1.52	< 0.01	9.09
50	< 0.01	0.09	0.83	< 0.01	15.51	0.17	193.28
100	< 0.01	0.36	4.49	0.30	165.84	1.59	4309.30
200	< 0.01	2.00	25.87	2.89	1946.63	21.01	—
400	< 0.01	32.61	164.57	29.74	26940.49	332.33	—
800	0.01	243.75	1410.96	309.68	—	5268.58	—

Design 2 (degenerate)		
$d_u(n_h)$	GRS	GK/RC
4 (6)	< 0.01	0.01
5 (10)	< 0.01	0.06
6 (15)	< 0.01	0.50
7 (21)	0.15	4.59
8 (28)	1.27	49.52
9 (36)	11.81	668.34
10 (45)	100.22	6803.53

Notes: Cases that were infeasible with 24 hours of computing are marked as —. The GK and RC algorithms are identical when $d_u > 2$.

7.4 Figure 3, Figure 4, and Algorithm 3: profiling and parameter equivalence

Read:

- Figure 3 shows sign-vector patterns for a two-player game example under different parameter values.

- Figure 4 illustrates projection of joint (u, θ) cells onto parameter space and construction of equivalence-class representatives.
- Algorithm 3 computes profiling hyperplanes, which determine boundaries of projected cells in parameter space.

Economic message: If different parameter values generate the same cell structure, the same linear programs do not need to be solved repeatedly. Profiling reduces redundant computation.

COUNTERFACTUAL IDENTIFICATION AND LATENT SPACE GENERATION

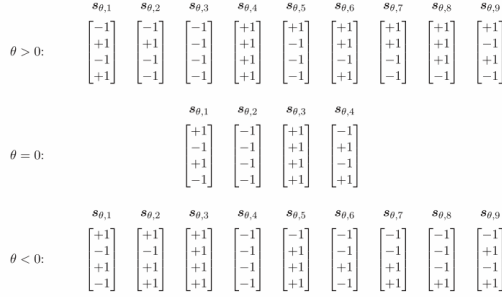


FIGURE 3

The sign vectors for the two-player game example when $\theta > 0$, $\theta = 0$, and $\theta < 0$.

Figure 3: sign vectors in the game example

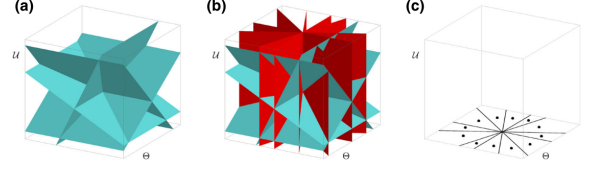


FIGURE 4

An illustration of the profiling procedure, starting with the initial arrangement in Figure 4(a), adding the profiling hyperplanes in Figure 4(b), and ending with the equivalence classes from Definition 3.1 illustrated with all representative points in Figure 4(c). (a) A hyperplane arrangement in $\mathbb{R}^d$. (b) The arrangement from 4(a), and all profiling hyperplanes and (c) The boundaries of the equivalence classes from Definition 3.1, and a representative point from each equivalence class.

Figure 4: profiling and equivalence classes

Algorithm 3: (Profiling Algorithm)

input : An $n \times d$ matrix A , with $d = d_u + d_\theta$ and $n, d_u, d_\theta \geq 1$ (with the last d_θ columns of A corresponding to θ).

output: A matrix V with d columns, the normal vectors for the profiling hyperplanes.

1. If any rows in A contain at least d_u leading zeros, include them as rows in V . Discard those rows from A and call the remaining matrix A^* .
2. **If** $\text{row}(A^*) \leq 1$, terminate and return V ; **else**: let $\mathcal{R}_2$ be the collection of all size-2 subsets of rows from A^* .
 - (a) Combine each pair of rows into a $2 \times d$ matrix and convert the matrix to reduced row echelon form.
 - (b) Check the rows of the matrix from 2(a) for non-zero vectors containing at least d_u leading zeros, and store these vectors in V .
 - (c) Store each pair of rows yielding a vector with at least d_u leading zeros in $\mathcal{R}_2^*$.
3. For $j = 3, \dots, d_u + 1$,
 - (a) **If** $\text{row}(A^*) \leq j - 1$, terminate and return V ; **else**: let $\mathcal{R}_j$ be the collection of all size- j subsets of rows in matrix A^* .
 - (b) Discard the subset $R \in \mathcal{R}_j$ if $R^* \subset R$ for any $R^* \in \mathcal{R}_k^*$ with $k < j$.
 - (c) **If** no subsets remain in $\mathcal{R}_j$ after step (b), set $\mathcal{R}_j^* = \emptyset$ and move to the next iteration; **else**:
 - (i) For all surviving subsets $R \in \mathcal{R}_j$, combine the elements of R into a $j \times d$ matrix, and convert the matrix to reduced row echelon form.
 - (ii) Check the rows of the matrix from step 3(c)(i) for non-zero vectors with at least d_u leading zeros, and store these vectors in V .
 - (iii) Store those size- j subsets yielding a non-zero vector with at least d_u leading zeros in $\mathcal{R}_j^*$, and proceed to the next iteration.

7.5 Tables 3–4: mobile phone application

Read:

- Table 3 describes the two mobile phone counterfactuals: reducing the price of phone 1 and eliminating phone 3 from the choice set.
- Table 4 reports bounds under combinations of assumptions, along with cell counts, computation time, and memory usage.
- For P1, the bounds tighten meaningfully as sign restrictions, fixed coefficients, and median-zero assumptions are imposed.
- For P2, the bounds remain at $[0.00, 0.29]$ across many specifications, showing the limited identifying power for the heterogeneous choice-set counterfactual.

Economic message: The mobile phone application demonstrates both the promise and the limits of nonparametric counterfactual analysis: price counterfactuals can be informative, but choice-set counterfactuals may need stronger assumptions.

TABLE 3
A description of the two counterfactuals considered for the phone choice example

Objective	Description	M_2	γ_1		γ_2	
			c_x	M_2^*	c_x	M_2^*
$P_{Y_1}(Y_{T1} = 1)$ $-P_{Y_1}(Y_{T2} = 1)$	Effect of reducing the price of phone 1	(1)	$c_x(P_1, P_2, P_3, W_1, W_2, W_3)$ $= (0.5, 2.5, 1.5, W_1, W_2, W_3)$	(1)	$c_x(P_1, P_2, P_3, W_1, W_2, W_3)$ $= (2.5, 2.5, 1.5, W_1, W_2, W_3)$	(1)
$P_{Y_1}(Y_{T1} = 1)$ $-P_{Y_1}(Y_{T2} = 1)$	Effect of eliminating phone 3 from the choice set	(1, 2, 3, 4)	$c_x(P_1, P_2, P_3, W_1, W_2, W_3)$ $= (P_1, P_2, P_3, W_1, W_2, W_3)$	(2)	$c_x(P_1, P_2, P_3, W_1, W_2, W_3)$ $= (P_1, P_2, P_3, W_1, W_2, W_3)$	(1)

Here M_2 is an index set for the choice sets faced by the consumer. We label the choice sets $\{0, 1, 2, 3\}, \{0, 1, 2\}, \{0, 1, 3\}, \{0, 2, 3\}$ with the indices 1, 2, 3, 4. For instance, $\pi(1)$ means the choice set is fixed at $\{0, 1, 2, 3\}$.

Table 3: phone choice counterfactuals

TABLE 4
Results for the phone choice application for the counterfactuals in Table 3

(MZ)	(FC)	(SR1)	(SR2)	RP ^a	Cells (Unstitched) ^b	(P1)	(P1) Cells (Stitched) ^b	Time (min) ^c	Memory (mb) ^d	(P2)	(P2) Cells (Stitched) ^b	Time (min) ^c	Memory (mb) ^d
					1,500,470	[-0.53, 0.86]	45,707	1.5	2,035	[0.00, 0.29]	373,642	4.7	4,167
					1,513,966	[-0.52, 0.86]	46,678	1.5	2,174	[0.00, 0.29]	378,362	3.6	4,317
					1,500,666	[0.00, 0.86]	45,721	1.5	2,151	[0.00, 0.29]	373,702	3.8	4,458
					1,516,188	[0.00, 0.86]	46,696	1.5	2,291	[0.00, 0.29]	378,432	3.3	4,811
				208	51,711	[-0.52, 0.85]	2,645	11.4	183	[0.00, 0.29]	14,548	35.1	270
				104	51,711	[-0.52, 0.85]	2,434	6.0	183	[0.00, 0.29]	14,127	17.7	270
				104	51,711	[0.00, 0.85]	2,118	6.3	183	[0.00, 0.29]	12,996	17.6	270
				52	51,711	[0.00, 0.85]	2,024	3.2	183	[0.00, 0.29]	12,906	8.8	270
					1,691,050	[-0.53, 0.50]	61,009	2.0	2,715	[0.00, 0.29]	446,443	6.5	5,834
					1,709,084	[-0.52, 0.50]	62,455	1.9	2,882	[0.00, 0.29]	452,474	4.8	6,258
					1,691,246	[0.00, 0.49]	61,046	1.8	2,848	[0.00, 0.29]	446,531	5.1	6,179
					1,709,304	[0.00, 0.48]	62,505	1.8	3,016	[0.00, 0.29]	452,578	4.2	6,608
				212	56,521	[0.00, 0.48]	3,111	14.2	204	[0.00, 0.29]	17,216	51.6	302
				106	56,521	[0.00, 0.48]	3,057	7.4	204	[0.00, 0.29]	16,598	26.4	302
				106	56,521	[0.00, 0.48]	2,789	7.7	205	[0.00, 0.29]	14,771	25.7	302
				53	56,521	[0.00, 0.48]	2,727	4.0	204	[0.00, 0.29]	15,321	13.0	302

ex: ^aNumber of representative points. ^bModal number of cells across all representative points. ^cTime required (for stitched cells). ^dMaximum memory required (for stitched cells).

Table 4: phone choice bounds and computation

7.6 Tables 5–6: airline entry application

Read:

- Table 5 describes four counterfactuals for the Kline and Tamer airline entry game.
- Table 6 reports bounds, cell counts, computation time, and memory use for those counterfactuals under different assumptions.
- The airline setting highlights the role of strategic interaction and equilibrium restrictions.
- Equilibrium assumptions can substantially alter the identified set for some counterfactuals while leaving others unchanged.

Economic message: The airline example shows that the framework handles games with multiple equilibria and demonstrates how equilibrium restrictions can be integrated into the same linear-programming framework.

TABLE 5
A description of the four counterfactuals considered for the *Kline and Tamer (2016)* entry game example

Objective	Description	γ_1		γ_2			
		c_x	y^*	c_x	y^*		
$P_{Y_1}(Y_{T1}^{LC} = 1)$ $-P_{Y_1}(Y_{T2}^{LC} = 1)$	ATE on Y_{T1}^{LC} of moving y^{TH} from 0 $\rightarrow$ 1	$c_x(M_1^{LC}, M_1^{TH}, S_1)$ $= (M_1^{LC}, M_1^{TH}, S_1)$	(-, 1)	(2)	$c_x(M_1^{LC}, M_1^{TH}, S_1)$ $= (M_1^{LC}, M_1^{TH}, S_1)$	(-, 0)	(2)
$P_{Y_1}(Y_{T1}^{TH} = 1)$ $-P_{Y_1}(Y_{T2}^{TH} = 1)$	ATE on Y_{T1}^{TH} of moving y^{TH} from 0 $\rightarrow$ 1	$c_x(M_1^{LC}, M_1^{TH}, S_1)$ $= (M_1^{LC}, M_1^{TH}, S_1)$	(1, -)	(1)	$c_x(M_1^{LC}, M_1^{TH}, S_1)$ $= (M_1^{LC}, M_1^{TH}, S_1)$	(0, -)	(1)
$P_{Y_1}(Y_{T1}^{LC} = 1)$ $-P_{Y_1}(Y_{T2}^{LC} = 1)$	ATE on Y_{T1}^{LC} of moving y^{TH} from 1 $\rightarrow$ 3	$c_x(M_1^{LC}, M_1^{TH}, S_1)$ $= (M_1^{LC}, 1, S_1)$	-	$\emptyset$	$c_x(M_1^{LC}, M_1^{TH}, S_1)$ $= (1, M_1^{TH}, S_1)$	-	$\emptyset$
$P_{Y_1}(Y_{T1}^{TH} = 1)$ $-P_{Y_1}(Y_{T2}^{TH} = 1)$	ATE on Y_{T1}^{TH} of moving y^{TH} from 1 $\rightarrow$ 3	$c_x(M_1^{LC}, M_1^{TH}, S_1)$ $= (1, M_1^{TH}, S_1)$	-	$\emptyset$	$c_x(M_1^{LC}, M_1^{TH}, S_1)$ $= (1, M_1^{TH}, S_1)$	-	$\emptyset$

Note when $\Pi^* = \emptyset$ the value of y^* does not matter, and so is marked with -.

Table 5: airline entry counterfactuals

TABLE 6
Results for the Kline and Tamer (2016) application for the counterfactuals in Table 5

(EQ)	(MZ)	(FC)	(SR1)	(SR2)	RP ^a	Cells (Unstitched) ^b	Cells (Stitched) ^b	Time (min) ^c	Memory (mb) ^d	(KT1)	(KT2)	(KT3)	(KT4)
					144	2,401	1,696	6.2	226	[-0.69, 0.28]	[-0.59, 0.18]	[-0.81, 0.82]	[-0.82, 0.85]
					144	3,136	2,800	3.8	258	[-0.69, 0.00]	[-0.59, -0.01]	[-0.63, 0.57]	[-0.59, 0.57]
					57,600	169	145	147	3,747	[-0.69, 0.23]	[-0.58, -0.01]	[-0.63, 0.51]	[-0.58, 0.50]
					14,400	169	151	42.6	1,304	[-0.69, 0.00]	[-0.58, -0.01]	[-0.63, 0.51]	[-0.58, 0.50]
					900	169	151	3.8	523	[-0.69, 0.00]	[-0.58, -0.01]	[-0.63, 0.51]	[-0.58, 0.50]
					196	3,136	2,839	12.3	304	[-0.69, 0.21]	[-0.59, 0.17]	[-0.81, 0.82]	[-0.82, 0.85]
					196	4,096	3,616	7.2	379	[-0.69, -0.30]	[-0.59, -0.01]	[-0.63, 0.57]	[-0.59, 0.57]
					9	4,096	2,977	0.5	146	[-0.69, -0.30]	[-0.59, -0.01]	[-0.63, 0.57]	[-0.59, 0.57]
					97,344	196	172	332.6	7,021	[-0.68, -0.37]	[-0.58, -0.02]	[-0.63, 0.51]	[-0.58, 0.50]
					24,336	196	178	90.6	2,318	[-0.68, -0.37]	[-0.58, -0.02]	[-0.63, 0.51]	[-0.58, 0.50]
					1,521	196	172	6.6	818	[-0.68, -0.37]	[-0.58, -0.02]	[-0.63, 0.51]	[-0.58, 0.50]
					144	2,401	2,191	7.2	588	[-0.69, 0.28]	[-0.59, 0.18]	[-0.81, 0.82]	[-0.82, 0.85]
					144	3,136	2,800	4.3	617	[-0.69, 0.00]	[-0.59, -0.01]	[-0.63, 0.51]	[-0.58, 0.50]
					9	3,136	2,800	0.5	493	[-0.69, 0.00]	[-0.59, -0.01]	[-0.63, 0.51]	[-0.58, 0.50]
					57,600	169	151	167.4	4,105	[-0.69, 0.23]	[-0.58, -0.01]	[-0.63, 0.51]	[-0.58, 0.50]
					14,400	169	151	6.662	1,662	[-0.69, 0.00]	[-0.58, -0.01]	[-0.63, 0.51]	[-0.58, 0.50]
					900	169	151	4.9	881	[-0.69, 0.00]	[-0.58, -0.01]	[-0.63, 0.51]	[-0.58, 0.50]
					196	3,136	2,839	14.4	306	[-0.69, 0.21]	[-0.59, 0.17]	[-0.81, 0.82]	[-0.82, 0.85]
					196	4,096	3,616	8.5	382	[-0.69, -0.30]	[-0.59, -0.01]	[-0.63, 0.51]	[-0.58, 0.50]
					9	4,096	2,977	0.6	149	[-0.69, -0.30]	[-0.59, -0.01]	[-0.63, 0.51]	[-0.58, 0.50]
					97,344	196	172	353.4	7,021	[-0.69, -0.37]	[-0.58, -0.02]	[-0.63, 0.51]	[-0.58, 0.50]
					24,336	196	178	102.6	2,318	[-0.69, -0.37]	[-0.58, -0.02]	[-0.63, 0.51]	[-0.58, 0.50]
					1,521	196	172	7.4	819	[-0.69, -0.37]	[-0.58, -0.02]	[-0.63, 0.51]	[-0.58, 0.50]
					N/A	N/A	N/A	N/A	N/A	[-0.52, -0.32]	[-0.50, -0.17]	[-0.39, 0.22]	[-0.45, 0.12]

ex: ^aNumber of representative points. ^bModal number of cells across all representative points. ^cTime required (for stitched cells). ^dMaximum memory required (for stitched cells).

Table 6: airline entry bounds and computation

8 Short summary

- The paper provides a unified framework for counterfactual identification in discrete outcome models with endogenous regressors and multidimensional latent variables.
- The central theoretical result is that, with discrete covariates and outcomes, an infinite-dimensional latent variable distribution can be replaced by a finite distribution over hyperplane-arrangement cells.
- Sharp bounds on counterfactual functionals can then be computed by linear programming.
- The paper introduces a fast exact cell enumeration algorithm for hyperplane arrangements.
- When the model is linear in parameters, profiling hyperplanes enumerate equivalence classes in parameter space, reducing redundant computation.
- Applications to mobile phone choice and airline entry illustrate how the method works in discrete choice models, heterogeneous choice-set models, and games with multiple equilibria.

- Additional assumptions such as independence, sign restrictions, median restrictions, and equilibrium restrictions can be added as linear constraints.
- The paper’s main practical contribution is to make sharp nonparametric counterfactual bounds computationally tractable in high-dimensional discrete outcome models.

9 Conclusion

9.1 Core conclusion

Gu, Russell, and Stringham show that counterfactual identification in a broad class of discrete outcome models can be reduced from an infinite-dimensional problem to a finite-dimensional one. The reduction is exact, not approximate, and it relies on the geometry of hyperplane arrangements induced by the model’s inequalities.

9.2 Economic intuition

For a discrete outcome model, the latent variable matters only through the signs of finitely many inequalities. Those signs are constant within each cell of the latent space. Therefore, all latent draws in the same cell are observationally equivalent for the purpose of matching observed data and generating counterfactual outcomes. Counterfactual bounds are then obtained by assigning probability mass to cells and choosing admissible selections.

9.3 Takeaway

The paper gives researchers a practical way to compute sharp bounds on counterfactuals in models that previously seemed computationally out of reach. Its broader message is that nonparametric identification and computation can be combined when the model’s discrete structure is exploited carefully.